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SUBJECT: CN-observation STD: DIV:

SCHOOL:

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Ex-no: 01

Date: 20/7/2026

Basic Networking Command

1. arp -a

Shaw IP-to-Mac address mappings. [arp -a]

o/p:- gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enpos3.

2. hostname

display the name of the computer

o/p: localhost.localdomain

3. ipconfig /all

Displays detailed network configuration

o/p: windows ip configuration.

Host name.....: Kaviya

Primary DNS suffix...:

Node Type....: Mixed

4. nbStat -a

Troubleshoot NetBios name resolution

o/p: Node Address: [192.168.1.100]

Name	Type	Status
Server 01	<00> unique	Registered
Server 02	<20> unique	Registered

5. o/p:

enpos2 : < BROADCAST, MULTICAST, DOWN, LOWER-UPS.

1. ip

Display IP address

Configure network interface

Add / delete IP address

Managing routing tables.

2. ip address add

Assigns a new IP address to a network interfaces

Ex: ip address add 192.168.1.254/24.

O/P: Lo: <Loop BACKUP, LOWER_UP>

3. Delete an IP Address

Remove the IP address from the interface

Eg: ip address del 192.168.1.254/24 dev exp0s3

O/P:

exp0s3: <BRODEAST, MULTICAST, UP, LOWER_UP>

4. Bring Interface online

Enables the network interface

E.g: ip link set exp0s3 down.

5. Interface online

Displays the network interface

E.g: ip link set exp0s3 down.

6. Enable promiscous Mode

Allows the interface receive all network Packet.

E.g: ip link set expos3 promisc on

o/p: expos3: <BROADCAST, MULTICAST, PROMISC>

7. Add a default route

Adds the default gateway for all network.

o/p:

default via 192.168.1.254 dev expos3.

8. Add a route through a Gateway:

Routes traffic for the 192.168.1.0/24 network through gateway 192.168.1.254

9. Delete a route

Delete the route to the 192.168.1 of 24 network through gateway 192.168.1.0.124

Eg: ip route delete 192.168.1.0/24 via 192.168.1.254

10. if Config

Configuring and Troubleshoot Network

o/p: Media State Media disconnected:

6. Enable promiscous Mode

Allows the interface receive all network Packet

E.g: ip link set expos3 promise on

O/P: expos3 : <BROADCAST3MULTICAST3 PROMISE>

7. Add a default route

Adds the default gateway for all network traffic -

O/P:
default Via 192.168.1.254 dev expos3

8. Add a route through a Gateway

Routes traffic for the 192.168.1.0.24 network through gateway 192.168.1.254

O/P: ip route add 192.168.1.0/24, dev etho

9. Delete a route

Delete the route to the 192.168.1.0/24, network through gateway 192.168.1.0.124.

E.g: ip route delete 192.168.1.0/24 via 192.168.1.254

10. If Config

Configuration and Troubleshoot Network

O/P: Media State . . . : Media disconnected.

11. mtr

combines the functionality of the ping and traceroute commands.

O/P: Host

1.172.16.8.1 Loss % 134 0.1 0.2 0.1

0.0

12. a) mtr

b) mtr - g google.com [shows numeric ip address]

c) mtr - b google.com [numeric ip address and hostnames]

d) mtr - c 10 google.com [numeric of ping that you have]

13. tcdump.

Designed for capturing and displaying packets

i) tcpdump -i eth0 -c to host 8.8.8.8

Filter our traffic coming from a specific host.

14. Ping

Display best way to test connectivity bt 2 nodes.

O/p:

Minimum = 0ms, Maximum = 0ms, Average = 0ms.

Result:

27/7/20

The Basic Networking and Linux command and learned and executed successfully.

Ex.no:02

DATE: 28/7/26

Different Type of Network cables

Aim:

Study of different types of Network cables.

a) Understand different types of Network Cables.:

1. Unshielded Twisted pair (UTP) cable.
2. Shielded Twisted pair (STP) cable
3. Coaxial cable.
4. Fibre optic cable

cable type	category	Max data Transmission	Adv/ Disadvantages	Application
UTP	Category 3	10bps	Adv: *Cheaperion cost *Easy to install	10Base-T Ethernet
	Category 5	up to 100MbPS		Fast Ethernet
	Category5e	1 GbPS	DIS: *Moderate to EMI	Fast ethernet
STP	Category 6,6a	10GbPS	Adv: *Shilded *Faster than UTP. *Less suspetible to noise	Gigabit Ethernet 10G Ethernet (55m) widely used in data centres

STP	Catagory 7	10 GbPS	Dis: * Expensive * Greater installation effort	Gigabit Ethernet, 10 GIE thernet (100m)
coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	* high bandwidth * Immune to interferance * Loss loss bandwidth. * Versatile. Disadvantages: * Limited distance * cost * Size is bulky.	Speed of signal is 500m Television network high speed internet connections.
Fibre optic	Single mode multi mode	100 GbPS	Adv: * high speed * high security * Long distance Dis: * Expensive * Requires skilled installers.	Maximum distance of fibre optics cable is around 80km

Result:

The different types of network cables are learned and executed successfully.

4/8/20

E.no :03

Date : 05/8/24 CISCO PACKET TRACER

Q) Analyse the following behaviour of network device using : CISCO PACKET TRACER simulator.

1. From the network component box, click and drag-and-drop the below components.

a. 4 Generic PCs and one HUB

b. 4 Generic PCs and One Switch.

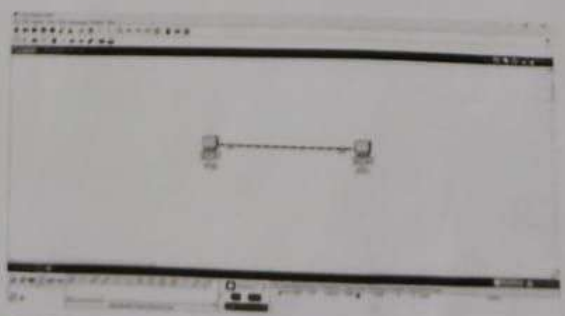
2. Click on Connection:

* click on Copper Straight-Through cable

* Select One of the pc and Connect it to HUB using the cable.

* Similarly connect 4 PC's to the Switch using copper Straight-through cable

3. Click on the PCs connected to hub, go to the desktop tab



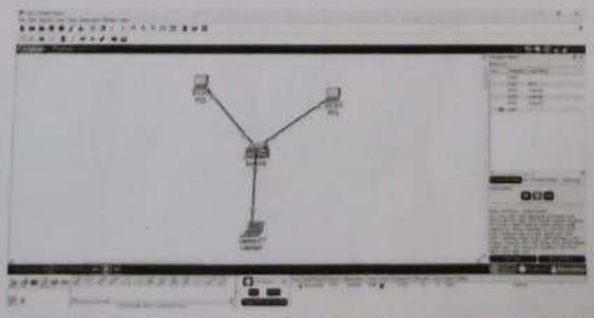
b. 4 Generic PCs and one Switch.

* click on copper straight - through cable
* Select one of the pc and connect it to HUB using the cable.
* Similarly connect 4 PCs to the switch using copper straight - through cable.

3. Click on the PCs connected to Switch, go to the desktop tab, click on IP configuration, and enter an IP address and subnet mask. Here the default gateway and DNS Server information is not needed as there are only two end device in the network.

4. Click on the PDU from the common tool bar.

a) Drag and drop it on the PC and it on another pc connected to the HUB.

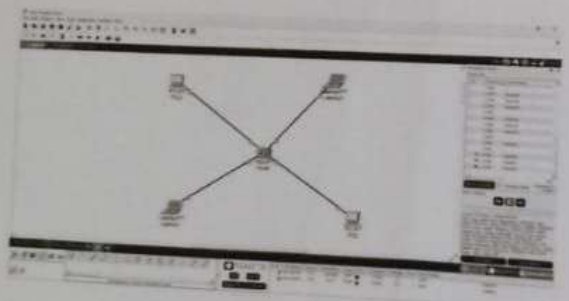


1. Connect PC1 and PC2 using an Ethernet cable.

2. Assign IP address to both PCs in the same network.

3. Check the connection using ping between PC1 and PC2.

4. If the ping succeeds, the PC-to-PC connection is established.



Result:

CISCO PACKET TRACER are learned and executed successfully

10/8/26

Ex.no: 01 Setting up a Local Area Network

Date 10.8.26

Aim:

To design, set up and configure a Local area Network connecting four computers through a network, switch, assign static IP address to all device.

Diagram:

LAN was implement using STAR Topology.

Implementation:

Step 1: Place the Device:

Open the CISCO packet Tracer

From the device category panel, select End Device and drag 4PCs onto the workspace.

Step 2: Cable the Device

Click connection → Select copper straight through cable.

Click each PC, then click the switch and Select an available FastEthernet port to complete to each connection.

Repeat for all four PCs Green link lights confirm a successfully physical connection.

Step 3: Assign IP address to PCs.

Click a PC → Desktop tab → IP configuration

Enter the state IP Address and submit mask as per the IP table for each PC.

Step 4:

Click a switch → CLI Tab

commands:

Enable

Configure Terminal

Interface Vlan 1

ip address 10.1.1.5 255.0.0.0

no shutdown

exit

Step 5:

Verify connectivity using ping

Check a pc → Desktop tab → Command prompt use the ping command to test connectivity to the other pc.

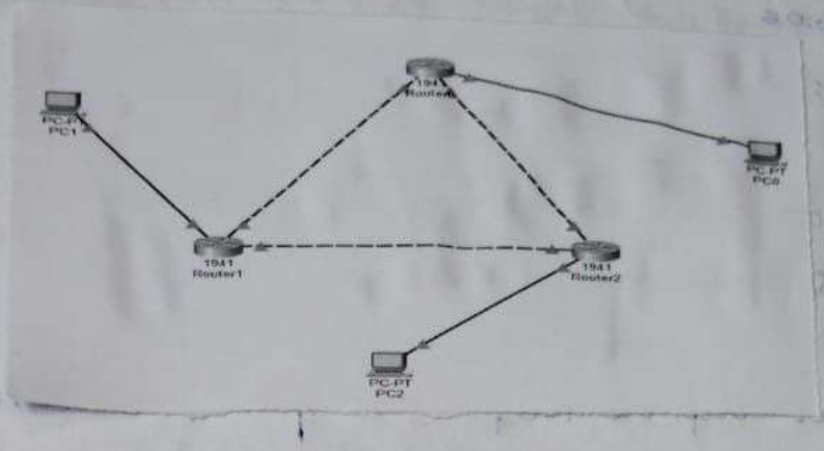
Step 6: Demonstrate data sharing.

Add a service device via PTs and connect PT to the switch.

Step 7: Observe packet flow in simulation mode.

Switch to simulation mode.

Send a ping or web request bet two devices. and use the play controls to step through the switch, as it travels hop by hop through the switch.



Result:

10/8/26 Setting up a Local Area Network (LAN) we learned and executed successfully.

Ex.no:05

Date: 18.8.26 Experiments on packet Capture tools: wireshark.

capturing and Analyzing packets using wireshark tool:

To filter, capture, view, packets in wireshark tool. capture 100 packets from ethernet: IEEE 802.3 LAN Interface and save it.

Procedure:

Select Local area network Connection in wireshark Go to capture option.

Select stop capture automatically after 100 packets.

Save the packet.

1. Create a filter to display only TCP/UDP packets inspect the packets and provide the flow graph.

Procedure:

Select LAN connection in wireshark.

Go to capture options.

Selects stop capture automatically after 100 packets.

then click the start capture.

Save the packet.

2. Create a filter to display only ARP packets and inspect the packets.

Procedure:

Go to capture option

Select stop capture automatically after 100 packets.

Then click start capture.

Search ARP packets in search bar

save the packets.

3. Create a file to display only DNS packets and provide the flow graph.

Procedure:

Go to capture option

Select stop capture automatically after 100 packets.

Then click start capture.

Search DNS packets in search bar.

To see the flow graph click statistics flow graph.

Result:

Capturing and analysing packets using wireless are studied and executed successfully.

29/9/20

Ex.no:06

Date:07.9.26 Error Detection and Error correction

Aim:

Write a program to implement error detection and correction using HAMMING code concept make a test run to input data stream and error correction features.

Sender:

*Input to sender file should be a txt of any length.

*Apply hamming code concepts on the binary data and add redundant bits to it.

Code:

```
data = input ("Enter a data:")
m = len (data)
r = 0
while 2**r < (m+r+1):
    r+ = 1
hamming = [0] * (m+r+1)
j = 0
for (i in range (1, m+r+1):
if p ((i-1)! = 0
hamming [i] = int (data [j])
j+ = 1
for p in range (3):
    pos = 2**p
    parity = 0
    for j in range (1, m+r+1):
        if j and pos! = 0
            parity ^= hamming [j]
```

Receiver:

Receiver program should read the input from channel file.

Apply hamming code on the binary data to check for error

code:

```
Receiver = input ("Enter the received hamming code")
n = len (receiver)
hamming = [0] * (n+1)
    hamming [i] = int (receives [n-i])
r=0
while 2**r < n+1:
    r+=1
error =0

for p in range (r):
    pos = 2**p
    parity=0
    for j in range (1, n+1):
        if j and pos != 0:
            parity = hamming [j]
    error += parity * pos.
if error != 0.
    print ("Error Detected", error)
else:
    print ("there is no Error")
```

```
for i in range (n, 0, -1):
    Print(hamming[i], end="")
Print()
```

Sample Input:

Data bits: 1011

Hamming code : 1100110

Receive detected at position : 6

correct Hamming code : 1100110

Result:

Error detection and error correction using hamming code are learned and executed successfully.

29/9/24

Exp: 07 Nmap to Discover live Hosts using
Date: 22-9-26 Nmap scans on the Tryhackme platform

Aim :-

To Discover live hosts using Nmap Scans (ARP, ICMP, TCP/UDP) on the Tryhackme platform Room link; https://tryhackme.com/room/nmap01.

Procedure:

Task 1 :- Introduction

* ARP Scan: This scan uses ARP requests to discover leve hosts.

* ICMP Scan: This scan uses ICMP requests to identify live hosts.

* TCP/UDP Ping Scan: This Scan Sends packets to TCP ports and UDP ports to determine live hosts.

Task 2: Subnetworks

Network Segment:- It is a collection of computers linked together through a common medium.

Subnetworks - It is usually the equivalent of one or more networks segments linked together and configured to utilize the same router in an

IP networks.

* Subnets with /16
* Subnets with /24

Task 3 :- Enumerating Targets.

We need to identify the targets we want to scan before we conduct any "action".

1) List

- <MACHINE-IP> <scanme.nmap.org>
- will scan 3 IP address.

2) range

- 10.11.12.15-20
- will scan 6 IP address

3) subnet

- <MACHINE-IP /30
- will scan 4 IP address

nmap -SL <TARGETS>

Task 4 :- Discovering like hosts.

1) ARP, from layer - commonly used for main-in-meddle-Attack.

2) ICMP from network layer - commonly used in DDOS attacks.

3) TCP from Transport layer - SYN flooding, TCP reset, TCP session hijacking attacks.

Task 5 : Nmap host Discovery using ARP:

1) When a privileged user tries to scan LAN, Nmap uses ARP requests.

2) nmap -PR -sn <TARGETS>

-PR specifies that you only want an ARP scan.

4) nmap -PR -sn <ATTACKBOX-IP>/24.

Task 6 :- Nmap host discovery using ICMP.

1) Add -PE option to utilize ICMP echo requests to discover live hosts.

2) nmap -PE -sn TARGET to check host is live.

3) -PP option instructs Nmap to use ICMP timestamp requests.

nmap -PP -sn TARGET.

4) -PM option can enable this scan to ICMP address mask requests.

Task 7 :- Nmap host discovery using TCP and UDP.

1) Nmap to use TCP SYN 'ping' use -ps followed by the port number, range, list or a combination of this options.

2) nmap -ps -sn TARGET check the TCP port is open.

3) TCP ACK ping - you must be running Nmap as a privileged user.

4) UDP ping - Sending a UDP packet to an open port is unlikely to result in a response.

5) nmap -PU -sn TARGET -

Task 8 :- Using Reverse-DNS lookup.

1) Nmap is to use reverse-DNS Internet hosts.

2) -R option to query the DNS server even for offline hosts.

Task 9 : Summary

ARP scan, ICMP Echo scan, ICMP scan, ICMP address mask SCAN, TCP SYN ping scan, TCP ACK ping SCAN, UDP ping Scan.

Result :

NMAP to discover live host using ~~#~~gethackme platform was successfully executed

29/9/22

Ex: no: 08 End to End communication at
Date: 28.09.26 Transport layer.

a)

Aim:-

Implement echo, client server using TCP/UDP packets.

Procedure:

* Create a socket using python socket programming.
* Specify the IP address and port number.
* Bind the server socket to the IP address and port.
* Make the server listen for client connection
* Send and receive messages between client and server.
* Display the received message.
* close the socket connection.

Code:

TCP-Server.

```
import socket
Server = socket.socket(socket.AF_INET, socket.SOCK_STREAM).
Server.bind("127.0.0.1", 5000))
```

```
Server.listen()
print("Server is waiting for connection...")
conn, addr = server.accept()
print("client connected")
while True:
    data = conn.recv(1024).decode()
    if not data:
        breaks
    print("client:", data)
    conn.send(data.encode())
    conn.close()
    server.close()
```

TCP - Receiver (client)

```
import socket
client = socket.socket(socket(socket.AF.INET,
                              socket.SOCK-STREAM)
client.socket(("127.0.0.1", 5000))
message = input("Enter Message:")
client.connect(("message.encode()))
reply = client.recv(1024).decode()
print("Server:", reply)
client.close()
```

Terminal 1 - Server

Python tcp_server.py

Server is waiting for connection...

client connected

client : Hello

Terminal 2 - Client

python tcp_client.py

enter message : Hello

server : Hello

TCP echo is working.

Result :

The TCP/UDP packets using echo client server socket program is executed successfully.

b)

Aim:

Implement chat client server using TCP packets.

code:

Chat - Server side

```
import socket
Server = socket.socket(socket.AF_INET,
        socket.SOCKET_STREAM)
Server.bind(("127.0.0.1", 6000))
Server.listen(1)
print("waiting for client...")
conn, addr = Server.accept()
print("client connected")
while True:
    message = conn.recv(1024).decode()
    if message.lower() == "bye":
    print("client ended chat")
    break
    print("client:", message)
    reply = input("Server:")
    conn.send(reply.encode())
    if reply.lower() == "bye":
conn.close()
Server.close()
```

Client Client side

```
import socket
client = socket.socket (socket.AF_INET, socket.SOCK_STREAM)
Client connect (("127.0.0.1", 6000))
Print ("connected to server")
while True:
    message = input ("client:")
    client.send (message.encode())
    if message.lower() == "bye":
        breaks
    reply = client.recv (1024).decode()
    Print ("Server:", reply)
    if reply.lower() == "bye":
        break
client.close()
```

Terminal 1

```
python chat_server.py
waiting for client....
client connected
client : Hello
server : Hi
client : How are you?
Server : Fine.
```

Terminal 2

```
Python chat_clientfy
connected to server
client : Hello
Server : Hi
Client How are you?
Server : Fine.
```

Result:

The chat programming using sockets by TCP client and Server was implemented successfully.

27/9/20.

Ex: 09
Date: 28/9/26 SLIDING WINDOW PROTOCOL

Aim:

Write a program to implement flow control at datalink layer using sliding window protocol. Simulate the flows from one node to another.

Code:

```
import time
window = int(input("Enter Window size:"))
message = input("Enter message:")
start = 0
while start < len(message)
print("SENDER")
end = start + window
if end > len(message):
    end = len(message)
#sender buffer
sender_buffer = [ ]
for i in range(start, end):
    frame = (i, message[i])
    sender_buffer.append(frame)
    print("Frame no:", i, "DATA", message[i])
print("\n Sender Buffer:")
print(sender_buffer)
```

```
#Delay
    Print ("\n Waiting for receiver ...")
    time.sleep (1)
#Receiver
    Print ("RECEIVER")
    expected = Start
    error = False
    for frame in sender_buffer:
    frame_no = frame [0]
    data = frame [1]
    Print ("Received Frame :", frame_no, "DATA :",
                                        data)
    if frame_no = expected:
        expected = expected + 1
    else:
        error = True
        break
#Receiver buffer
    if error:
        receiver_buffer = ("NAck", expected)
        Print ("\n Receiver Buffer :")
        Print (receiver_buffer)
    else:
        ack = expected - 1
        receiver_buffer = ("ACK", ack)
        Print ("\n Receiver buffer :")
        Print ("Receiver_buffer")
```

```
# Sender reads ACK/NACK
Print ("\n SENDER: CHECKING RESPONSE")
response = receiver_buffer[0]
number = receiver_buffer[1],
if respone == "ACK":
    Print ("ACK received:", number)
    start = number + 1
else:
    Print ("NACK recived:", number)
    start = number
Print ("TRASMISSION COMPLETED")
```

output:

Ener Window size: 3

Enter message: Hello

SENDER

Frame no: 0 DATA: H

Frameno: 1 DATA: e

Frameno: 3 DATA: l

Sender buffer:

[(0,'H'), (1,'e'), (2,'l')] waiting for receiver...

waiting for receiver...

SENDER

Frameno: 3 DATA: l

Frameno: 4 Data: o

Sender Buffer:

[(3,'l'), (4,'o')]

RECEIVER

Received Frame: 0 DATA: H

Received Frame: 1 DATA: e

Received Frame: 2 DATA: l

Received Buffer:

('ACK', 2)

SENDER CHECKING RESPONSE

ACK Received: 2

RECEIVER

Received frame: 3 DATA: l

Received Frame: 4 DATA: O

Received Buffer:

('ACK', 4)

SENDER CHECKING RESPONSE

ACK Received: 4

Transmission Completed.

Result:

Sliding window protocols are learned and executed successfully.

8/9/24

Ex no: 10 Subnetting in cisco packet Tracer

Date: 22/9/26

Aim:

To Implement classless and subnetting using cisco packet tracer and verify the connectivity between different subnets.

Requirements:

cisco packet Tracer

Router

Switches

PCs

Ethernet Cables.

Procedure:

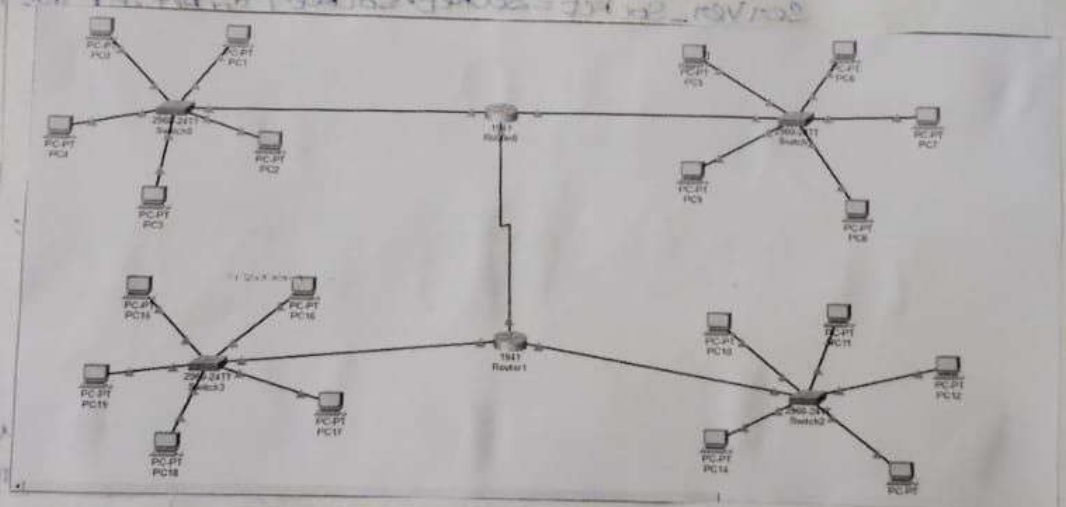
* Open cisco packet tracer.
* Create a new network topology.
* Add routers, switches and pcs to the workspace connect the device using ethernet cables.
* Divide the network into smaller subnets using the 127 subnet mark.
* Configure the Ip address, subnet mark and default gateway for each pc.
* Configure the required switch ports
* Enable the router interfaces using the no shutdown command.

* check the Ip configuration of all devices use the ping command to check connectivity between pcs.

* Verify that the devices in the network communicate successfully.

Ip Addressing :-

	Ip address	Subnet mask	Default Gateway
PC-PT-PC0	192.168.2.12	255.255.255.192	192.168.3.1
PC-PT-PC1	192.168.1.12	255.255.255.192	192.168.3.1
PC-PT-PC2	192.168.1.12	255.255.255.192	192.168.1.1
PC-PT-PC3	192.168.1.11	255.255.255.192	192.168.1.1



Result:

Thus, classless IP subnetting was successfully implemented using cisco packet tracer and connectivity between the network devices was verified successfully.

22/9/21

Ex.no:11

Date: 28/9/26 PING Program

Aim:

To implement a ping program using socket Programing to check the connectivity between two network devices.

Program:

```
import socket
import time
import sys
def udp_ping_server(host = "127.0.0.1",
                    port=12000)
server_socket=socket.socket(socket.AF_INET,
                            socket.SOCK_DGRAM)
server_socket.bind(host,pop)
Print(f"[server] listening on {host}:
                    {port}")
while True:
    msg,client address = server_socket.rev
                                from(1024)
    Print(f"[Server] Received' {msg.decode}
                from {client address}")
    Server.Socket.send to(msg, client_addr
def udp.ping.client(Server_host = "127.0.0.1",
                    12000, count = 5);
client_socket.settimeout(1)
```

```
for i in range(1, count+1):
msg = f"ping {i} {time.time()}"
start = time.time()
client_socket = sendto()
    msg.encode(), (server_host, server_port)
try:
    data = client_socket.recvfrom(1024)
    end = time.time()
    rett = (end-start) * 1000
    Print (
        f"Reply from {serverver_host} {server_
            port} {data.decode()}
    RTT = {rtt = 2f} ms')
except socket.timeout:
    Print (f"Request {i} timed out")
client_socket = close()
if __name__ == "__main__":
    if len(sys.arg) > 1 and sys.argv[1] == "serva":
        udp_ping_server()
    else
        udp_ping_client()
```

output :-

```
python udp_ping.py server
[SERVER] listening on 127.0.0.1:12000
[SERVER] Received ping 1791473643,6564088
                    from (1,2-7.0.0.1)
python udp_ping.py.
```

Reply from 127.0.0.1:12000 ping 179.47bat3:
=0.92ms

Reply from 127.0.0.1:12000 ping2 179c47b487.
6982551277 =0.25ms

Result:
This, the ping program is executed
Succesfully.

22/9/26

Ex.no:12

Date: 29/9/25 Implementation of Subnetting is CISCO Packet Tracer Simulator.

Aim:

To implement the subnetting in CISCO PACKET TRACER Simulator.

Procedure:-

* Open cisco packet tracer and create a new network topology.

* Add the required routers, switches and PCs to the workspace.

* Connect the device using appropriate copper cables.

* Use the network 192.168.1.0/24 and divide it into 127 networkes.

* Use enable and configure terminal in the router CLI.

* Enter the interface configuration terminal in the router and assign the Ip using:

(i) ip address <IP address> 255.255.255.254

(ii) no shutdown.

* Repeat the configuration for all required Subnets.

*Use ping command to test connectivity between pcs and the router.

* If ping is successfully, the network is configured correctly.

Result:

The network was successfully subnetted using the 127 Subnet mask in cisco packet tracer. connectivity between the devices was successfully verified using ping command.

Exno: 13

Date: 5/9/26 Routing at Network Layer

a) Simulate Static routing protocol configuration using cisco packet tracer.

Aim:

To Simulate a Static routing Configuration using cisco packet Tracer.

Procedure:

* Open cisco packet tracer and create the given network topology.
* Connect 3 routers and pcs using suitable cables
* Configure IP address for routers and pcs.
* Enable all router interfaces using no shutdown.
* Configure static routers on Router 0, Router 1 and Router 2 using Ip mobile.
* use Tracert to verify the routing path.
* Disconnect, the main link and verify that the backup route works.
* Thus a Static routing is Successfully configured and verified.

```
Router>enable
Router#show ip route static
S    30.0.0.0/8 [20/0] via 40.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2
S    60.0.0.0/8 [1/0] via 40.0.0.2

Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router#show ip route static
S    30.0.0.0/8 [10/0] via 20.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2
S    60.0.0.0/8 [1/0] via 40.0.0.2
```

Result:

Thus, Static routing was successfully configured using cisco packet Tracer. connectivity between all networks was verified successfully using ping.

b. Simulate Rip using CISCO packet Tracer

Aim :-

To Simulate Rip using CISCO packet Tracer.

Procedure:

* Create the topology with routers and 2pcs.
* Connect the device using suitable cables.
* Assign the given Ip address to both pcs.
* Configure Fastethernet and Serial interface of all routers.
* Enable interface using no shutdown.
* Let clock rate and Hoop and bandwidth to on DE Serial interface.
* Enable RIP on all router 0-Router 2 link and use tracer again to verify RIP selects the alternate route.

```
Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=55ms TTL=125
Reply from 20.0.0.2: bytes=32 time=56ms TTL=125
Reply from 20.0.0.2: bytes=32 time=60ms TTL=125
Reply from 20.0.0.2: bytes=32 time=55ms TTL=125

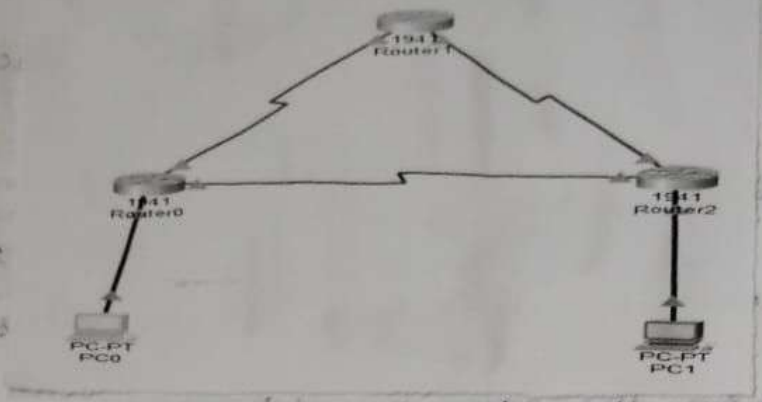
Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 55ms, Maximum = 60ms, Average = 56ms

C:\>tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    11 ms     8 ms      8 ms      192.168.1.250
  3    29 ms     26 ms     25 ms     192.168.1.245
  4    22 ms     28 ms     20 ms     20.0.0.2

Trace complete.
```



Result:

RIP successfully configured and dynamic routing between PC0 and PC1 was verified.

5/10/26

Ex.no 14

Date: 6/9/26 Interworking with routers CISCO PACKET TRACER

a) Design and configure a Simple interwork using router.

Aim:

To design and configure a simple internetwork using a router in CISCO PACKET TRACER and test connectivity between two PCs.

Procedure:

Step 1: Configuring Router.

* Select the router and open [CL]
* press enter to activate the previleged mode

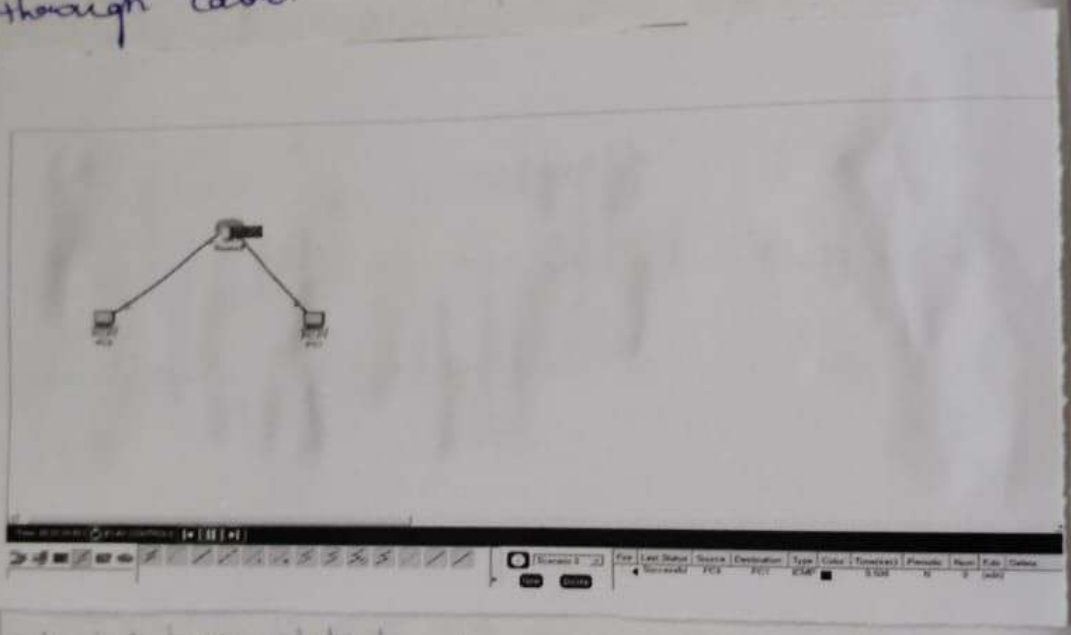
Step 2:

* Assign Ip address to every pc in the network.
* Select the PC, Go to the desktop and select Ip configuration and assign an Ip address, Default gateway subnet mask.
* Assign the default gateway of pc0 as 192.168.10.1
* Assign the default gateway of pc1 as 192.168.20.1

Step 3:

* Connect fast Ethernet 0 port of PC0 with Fast ethernet 0/0 port of Router 1 using a copper straight-through cable.

* Connect Fast Ethernet 0 port of PC1 with fast Ethernet 0/1 port of Router 1 using a copper straight through cable.



Result:

The Simple Internetwork was successfully designed and configured using a router connectivity between PC0 and PC1 was successfully verified using PDU.

b) Design and configure an internetwork using wireless router DHCP server and internet cloud.

Aim :

To Design and configure an internetwork using a wireless router, DHCP server and internet cloud in cisco packet Tracer.

Procedure :

* Open cisco packet Tracer and create the required network topology.

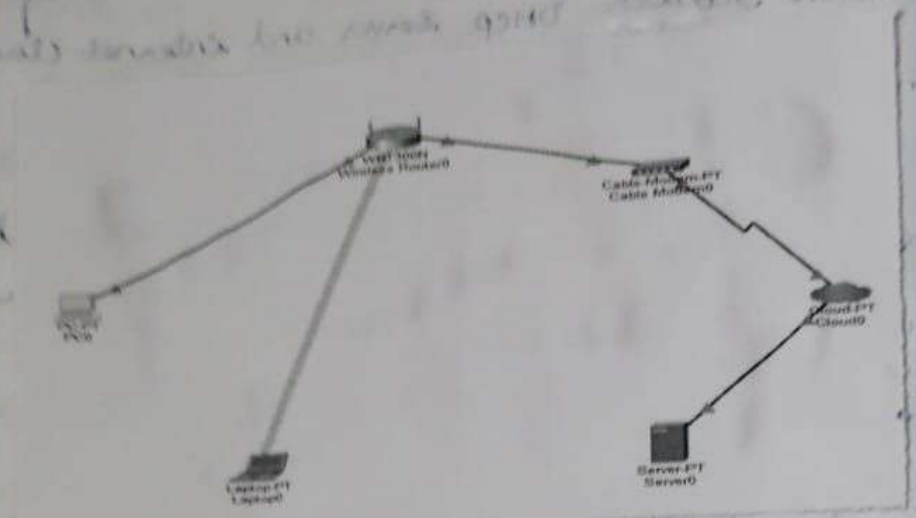
* connect PC, laptop, wireless router, cable modem, internet cloud and server.

* Configure the wireless router with SSID Homenetwork and enable DHCP.

* Connect the laptop to the wireless network and set the pc to DHCP.

* Configure the server with IP 208.67.220.220 and enable DHCP and DNS.

* Verify connectivity using ip config and ping cisco.com.



Result:

The Internetwork was successfully designed and configured using a wireless router, DHCP Server and Internet cloud and connectivity was successfully Verified using ping Cisco.com.

Ex. no

Date:

Aim:

To design and configure an Internetwork using a switch, router and ethernet cables and verify connectivity between computers.

Procedure:

* Open Cisco Packet Tracer and place end devices, a switch and router in the workspace.
* Connect the computers to the switch using ethernet cables.
* Connect the switch to the router using an ethernet cable.
* Configure the router interfaces with appropriate IP address and enable the interfaces.
* Use the ping command to test connectivity between computers.
* Observe the successful transmission of packets.

Result:

Thus, an Internetwork was successfully designed and configured using a switch, router and ethernet cables in Cisco packet Tracer.

Ex. no: 15

Date: 6/9/24 Analyze various types of servers using webalizer.

Aim:

To analyze web server log files using the webalizer tool and study server traffic statistics.

Procedure:

*Install and configure the webalizer tool on a system.

*Obtain the web server log file, such as an Apache access log.

* Configure webalizer to read the server log file.

* Execute the webalizer command to generate statistical reports.

*Open the generated HTML report in a web browser.

* Analyze the number of visitors, reports, page views, hits and bandwidth usage.

* Compare server activity using the generated reports.

Result:

Thus web server log files were analyze
Successfully using the webaliyer tool, and server
traffic statistics were studied.